

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We realize the generalized-baker benchmark as an ordinary physical piecewise-exact symplectic impact system and keep its coupled symbolic natural extension as a coding object rather than a fictitious symplectic inverse limit. A global affine work coordinate produces the additive current, while a refinement-independent graded flag-current complex gives typed all-depth parameter response. No bilateral Koopman gap or symbolic Hamiltonian leaf is used.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: THE PHYSICAL IMPACT SUSPENSION AND THE COMPLETE FLAG COMPLEX

The physical symplectic section and the symbolic natural extension are no longer identified topologically. The former is the ordinary square with a finite piecewise-smooth impact structure; the latter is a zero-dimensional coding

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space. This separation removes the false leaf claim while preserving an exact mechanical realization, the work cocycle, and the full response calculus.

1.1. Physical and symbolic objects. Let

$$Q = \mathbb{T}^2, \quad D_i(a) = I_i(a) \times \mathbb{T}, \quad H_i(a) = \mathbb{T} \times I_i(a),$$

and on the regular part of $D_i(a)$ put

$$B_{a,i}(q, p) = \left(\frac{q - s_{i-1}(a)}{p_i(a)}, s_{i-1}(a) + p_i(a)p \right).$$

The branch map is exact symplectic and maps $D_i(a)$ bijectively onto $H_i(a)$. The current vertical seam set and its forward/backward images are denoted by $\mathcal{S}_a$.

Let $\Sigma^+ = \mathcal{A}^{\mathbb{N}_0}$ and $\Sigma^- = \mathcal{A}^{\mathbb{N}}$. The natural-extension map is

$$\widehat{\sigma}(x, y) = (\sigma x, x_0 y).$$

Theorem 1.1 (Correct physical–symbolic factor). *There is a measurable coding map $\mathbf{c}_a : Q \setminus \mathcal{S}_a \rightarrow \Sigma^+ \times \Sigma^-$ such that*

$$\mathbf{c}_a \circ B_a = \widehat{\sigma} \circ \mathbf{c}_a.$$

It is one-to-one modulo the complete seam orbit and sends Lebesgue measure to the Bernoulli natural-extension law with weights $p_i(a)$. The symbolic space is used only as a coding space; no tangent bundle or symplectic form is assigned to it.

Proof. The future word records the successive vertical strips. The past word is read from the successive horizontal strips of the inverse branches. After one application of B_a , the used future symbol becomes the newest past symbol, which is exactly $(x, y) \mapsto (\sigma x, x_0 y)$. A finite two-sided cylinder pulls back to a rectangle of area equal to the product of its branch weights. Monotone convergence over cylinders proves the measure statement. Only points on an iterated seam have more than one coding germ. $\square$

1.2. An exact autonomous Hamiltonian impact system. For each branch let $S_{a,i}$ be the type-two generating function

$$S_{a,i}(q, P) = \frac{(q - s_{i-1}(a))(P - s_{i-1}(a))}{p_i(a)}.$$

Let $W_i = D_i(a) \times [0, 1]_\tau \times \mathbb{R}_E$ with symplectic form $dq \wedge dp + d\tau \wedge dE$. The incoming and outgoing ports are kept as separate boundary faces. At the outgoing face the canonical impact relation is the graph of $B_{a,i}$, which is exact Lagrangian because $B_{a,i}^*(p dq) - p dq = dS_{a,i}$.

Theorem 1.2 (Autonomous Hamiltonian impact realization). *There is a finite-dimensional autonomous Hamiltonian impact system $(\mathcal{W}_a, \Omega_a, \mathcal{H}_a)$ with the following properties.*

- (i) *Between ports the flow is smooth Hamiltonian and satisfies $\dot{\tau} = 1$.*

- (ii) Every impact is the canonical exact relation generated by $S_{a,i}$; no stochastic reset is added.
- (iii) The regular first-return map on the physical section Q is exactly B_a , with return time one.
- (iv) The Liouville incoming flux of port i is $p_i(a)$.

The impact system is a smooth symplectic manifold with boundary on each flight chamber and a standard exact symplectic impact relation at the finite seam set. No claim of a smooth flow through a discontinuity is made or needed.

Proof. The affine exact map $B_{a,i}$ is the time-one map of a quadratic Hamiltonian after the two canonical translations encoded by $S_{a,i}$. Autonomization adds the clock pair (τ, E) and the Hamiltonian $E + K_{a,i}(\tau, q, p)$. The branch chambers are disjoint before impact, and the outgoing exact Lagrangian graph identifies their flux with the common physical section. Standard composition of exact canonical relations gives a deterministic symplectic impact flow. Since $\dot{\tau} = 1$, every regular flight returns after one unit, and the generating equations give precisely $B_{a,i}$. The port flux is its q -width. $\square$

1.3. Canonical work family. Fix nonconstant branch works c_i . Starting from one reference weight vector p^0 , define the canonical family

$$p_i(\vartheta) = \frac{p_i^0 e^{\vartheta c_i}}{\sum_j p_j^0 e^{\vartheta c_j}}, \quad a(\vartheta) = \sum_i c_i p_i(\vartheta).$$

On every compact interval on which $\text{Var}_{p(\vartheta)} c > 0$, the map $\vartheta \mapsto a$ is a diffeomorphism. The parameter a used below is this mean-work parametrization.

Proposition 1.3 (Exact work holonomy and closure). *Attach to the i th chamber the affine work coordinate $u(1) = u(0) + c_i$. Its n -return holonomy is $A_n = \sum_{k \leq n} c_{x_k}$. Canonical tilting of the reference path law by $\exp\{\vartheta A_n\}$ gives exactly the member $p(\vartheta)$ of the same family, and the conjugate field is $\vartheta = I'(a)$.*

Proof. The affine coordinate adds once at every exact impact, so the holonomy is the sum of the branch works. The Radon–Nikodym derivative factorizes into

$$\prod_{k \leq n} \frac{p_{x_k}(\vartheta)}{p_{x_k}^0} = \exp\{\vartheta A_n - n\Lambda(\vartheta)\}.$$

Strict positivity of the work variance gives the Legendre inverse $\vartheta = I'(a)$. $\square$

1.4. The refinement-independent flag-current complex. Let $\mathcal{P}_a^{m,n}$ be the finite physical partition generated by seams at times $-m, \dots, n$. A presentation of a current consists of a smooth bulk density on every cell and a smooth density on every oriented flag of codimension $k \leq r$. Two presentations are equivalent when they agree on a common refinement after

cancellation of oppositely oriented interior flags. For the equivalence class $[U]$ define

$$\|[U]\|_{r;\rho_-,\rho_+,\kappa} := \inf_{U' \sim U} \sum_{C \in \mathcal{P}_a^{m,n}} \rho_-^m \rho_+^n \sum_{k=0}^r \kappa^k \sum_{F \in \mathfrak{F}_k(C)} \|U'_{C,F}\|_{C^{r-k}}.$$

Choose $4\rho_- < 1$, $4\rho_+ < 1$, and $M\kappa < 1$, where M bounds the number of new incident faces created in one refinement.

Theorem 1.4 (Complete graded flag complex). *The Hausdorff completion of the quotient above is a Banach complex $\mathfrak{C}_a^r$. Common refinement, the physical push-forward by B_a , the flag boundary, and parameter transport are bounded on the indicated levels. More precisely,*

$$D_a : \mathfrak{C}_a^r \longrightarrow \mathfrak{C}_a^{r-1}$$

after one derivative, and $D_a^q : \mathfrak{C}_a^r \rightarrow \mathfrak{C}_a^{r-q}$ contains exactly the flags of codimension at most q . The definition is independent of past/future presentation because the norm is an infimum over all common refinements.

Proof. Refining one future or past level creates at most four descendants, while a codimension- k flag creates at most M incident flags. The three geometric weights therefore make every refinement map summable. The infimum is a seminorm on the algebraic quotient; a zero-norm class has zero pairing with every compatible test form and is therefore zero. This proves Hausdorffness. The coupled shift changes (m, n) by at most one and hence has bounded norm with factor depending only on ρ_-/ρ_+ . Reynolds' formula on each polyhedron gives a bulk derivative plus its oriented face current. Iteration gives the standard chain of corner and higher flag terms, all already present in the complex. Closedness follows by completing the graph of the boundary operator. $\square$

1.5. All-order physical response. A transported observable is part of the theorem data: it is a C^r family F_a represented by a section of $\mathfrak{C}_a^r$. Let P_a^+ be the future Ruelle operator, P_a^- the conditional past operator, and $J_a f(x, y) = f(\sigma x, x_0 y)$ the present-symbol coupling.

Theorem 1.5 (Typed response and cancellation). *For every fixed $q < r$, the derivative $\partial_a^q \int_Q F_a dq dp$ exists. It is the absolutely convergent sum of Bernoulli score insertions, the reduced one-sided resolvents joined by J_a , and the bulk/flag Hadamard insertions from Theorem 1.4. Mixed future-past cylinders are evaluated with the projective tensor norm and are not replaced by independent dynamics. If F_a is one fixed physical function under the identity transport, the whole sum is zero.*

Proof. On a finite two-sided refinement, differentiation is ordinary finite calculus. Conditioning successively on future and past gives the two one-sided Poisson equations, while J_a inserts the common present symbol. Reynolds'

theorem gives all flag terms. The one-sided spectral gaps sum time insertions, and the three geometric weights sum the refinement and flag insertions. Dominated convergence passes to the completed complex. For an identity-transported physical function, Lebesgue measure is independent of a ; differentiating its change-of-variables identity proves exact cancellation. $\square$

Theorem 1.6 (Round-thirteen A1 closure). *The generalized baker family admits an exact autonomous Hamiltonian impact realization on the ordinary physical symplectic section, a correctly coupled symbolic natural extension, a canonically closed work family, and a refinement-independent all-order flag-current response calculus. No zero-dimensional symbolic inverse limit is presented as a symplectic leaf.*

Proof. Combine Theorems 1.1, 1.2, 1.4, and 1.5, with Proposition 1.3. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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